بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

PRESENTATION#01

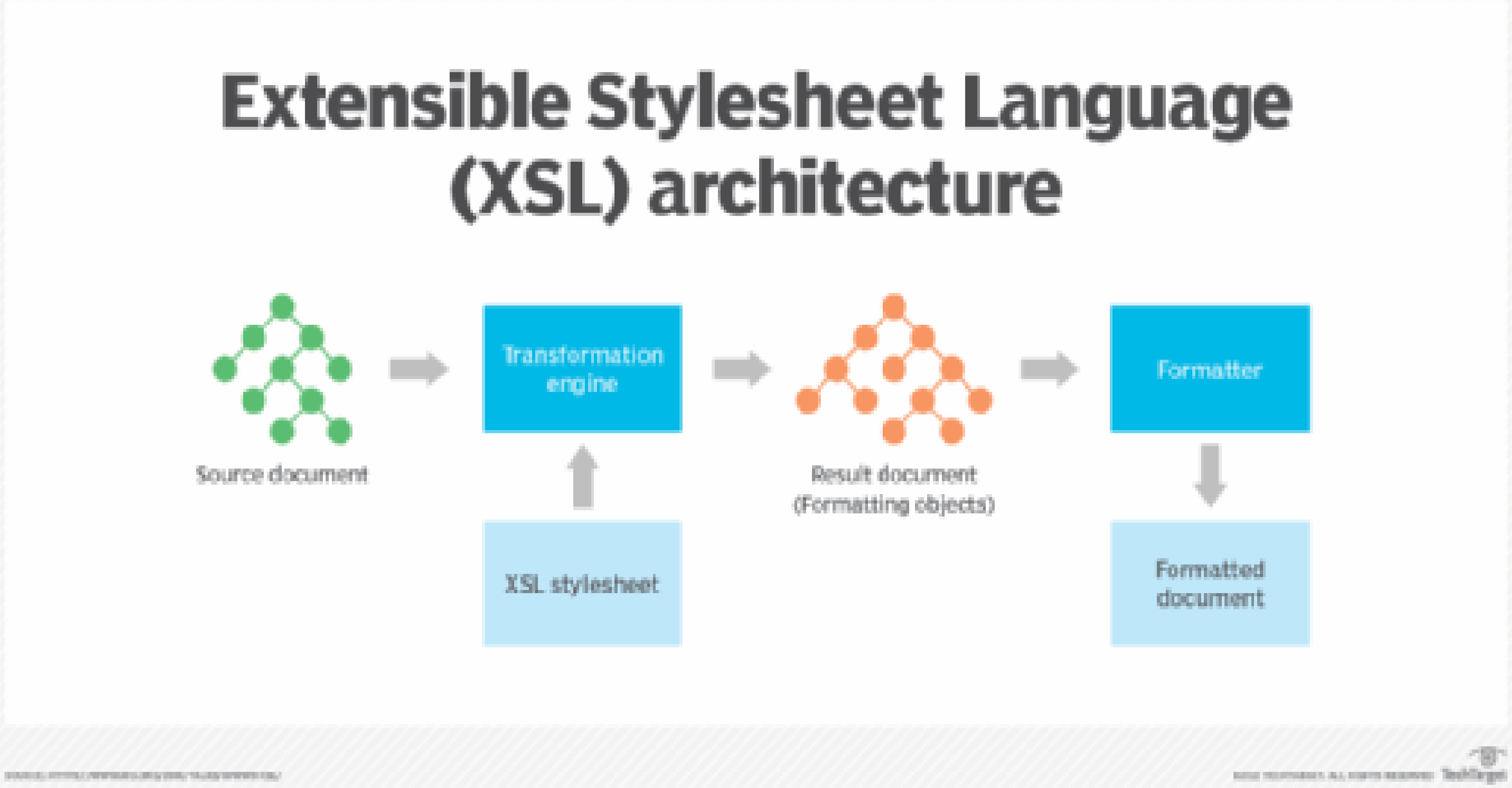
SUBMITTED BY: GROUP #03

Web services XSL

- XSLT
- XML-PATH
- XSL-FO

INTRODUCTION TO XSL

- XSL stands for Extensible Stylesheet Language
- XSL is the bridge between XML and HTML
- XSL was designed specifically to style XML pages and is much more sophisticated than CSS
- We can use XSL to have different HTML formats for the same data represented in XML
- Example : for example the text of a heading might also appear in a dynamically generated table of contents


Extensible Stylesheet Language (XSL) architecture
Source document
Transformation engine
XSL stylesheet
Result document (Formatting objects)
Formatter
Formatted document
TechTarget

Declaration/Syntax

```
<?xml version="1.0" encoding="UTF-8"?>
 <xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform" version="1.0">
 <!-- XSLT code goes here -->
 </xsl:stylesheet>
```

ELEMENTS OF XSL

1. **Templates:**

XSLT operates by matching patterns in the XML document and applying templates to them. Templates are defined using the <xsl:template> element.

```
<xsl:template match="...">
  <!-- XSLT instructions -->
</xsl:template>
```

2. **Match Patterns:**

Match patterns define the elements or nodes to which a template should apply. Patterns can use XPath expressions to specify the selection criteria.

```
<xsl:template match="book">
  <!-- Template for processing book elements -->
</xsl:template>
```

3. **Output Specification:**

The <xsl:output> element specifies the output format and encoding of the transformed result.

```
<xsl:output method="html" encoding="UTF-8"/>
```

4. **Built-in Functions:**

XSLT provides a set of built-in functions for string manipulation, node selection, arithmetic operations, etc. For instance:

```
<xsl:value-of select="concat('Hello ', $name)"/>
```

Code/Example

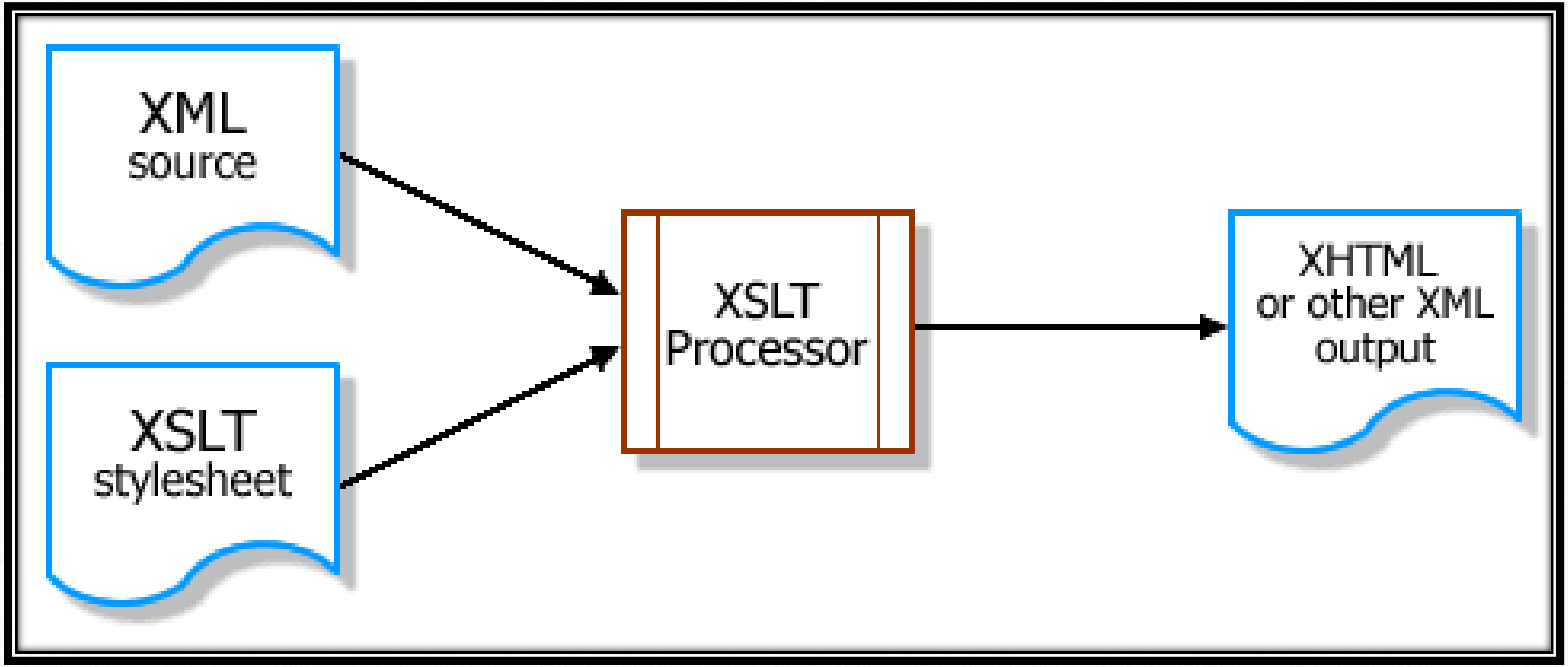
```
<?xml version="1.0"?><xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
<xsl:output method="text"/>
 <xsl:template match="/">
Article - <xsl:value-of select="/Article/Title"/>    Authors: <xsl:apply-
templatesselect="/Article/Authors/Author"/>
</xsl:template>
 <xsl:template match="Author">
 - <xsl:value-of select="." />
 </xsl:template>
</xsl:stylesheet>
```

Features of XSL

- Support for browsing, printing, and aural rendering.
- Formatting highly structured documents (XML)
- Performing complex publishing tasks: table of contents, indexes, reports
- Addressing accessibility and internationalization issues

Introduction to XSLT

- XSLT stands for Extensible Stylesheet Language for Transformation
- XSLT provide the mechanism for converting an XML document to other format
- The stylesheet contains conversion rules for accessing and transforming the input XML document to a different output format
- An XSLT processor is responsible for applying the rules defined in the style sheet to the input XML document


XML
source
XSLT
stylesheet
XSLT
Processor
XHTML
or other XML
output

Declaration

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform" version="1.0">
<!-- XSLT code goes here -->
</xsl:stylesheet>
```

Elements

XSLT Template: The basic building block of XSLT is the template. Templates define rules for transforming elements in the XML document.

```
<xsl:template match="pattern">
  <!-- Transformation instructions -->
</xsl:template>
```

XSLT Element:

The <xsl:element> element allows you to create elements in the output document.

```
<xsl:element name="element_name">
    <!-- Content -->
</xsl:element>
```

XSLT Attribute:

The <xsl:attribute> element allows you to create attributes for elements in the output document.

```
<xsl:attribute name="attribute_name">
    <!-- Attribute value -->
</xsl:attribute>
```

XSLT For-each:

The <xsl:for-each> element allows you to iterate over a selected set of nodes.

```
<xsl:for-each select="xpath_expression">
    <!-- Transformation instructions -->
</xsl:for-each>
```

Code/Example

```
<?xml version="1.0"?><?xml-stylesheet type="text/xsl" href="example.xsl"?>
<Article>
<Title>My Article</Title>
<Authors>
<Author>Mr. Foo</Author>
<Author>Mr. Bar</Author>
  </Authors>
 <Body>This is my article text.
</Body>
</Article>
```

Features of XSLT

- XSLT brings XML, schemes, and X-Path together in a declarative programming language
- It is used to query and transform XML(and with XSLT3, JSON) data, enabling one to one express data in new ways, or to create new data based on the content or structure of existing data

Introduction to X-Path

- X-Path stands for XML(Extensible Markup language) Path Language
- X-Path uses “path like” syntax to identify and navigate nodes in an XML document
- X-Path is major element in the XSLT standard
- X-Path contains over 200 built-in functions

Approaches of X-Path

Absolute XPath:

- Absolute XPath expressions provide the full path from the root element to the target element.
- They begin with a single forward slash (/) representing the root node of the document.
- Absolute XPath expressions are less resilient to changes in the document structure. If the structure of the document changes (e.g., elements are added or removed), absolute XPath expressions might become invalid.

Relative XPath:

- Relative XPath expressions specify the path from the current node to the target node.
- They do not necessarily start from the root node and are more context-dependent.
- Relative XPath expressions are more flexible and easier to maintain because they are based on the structure relative to a given context.

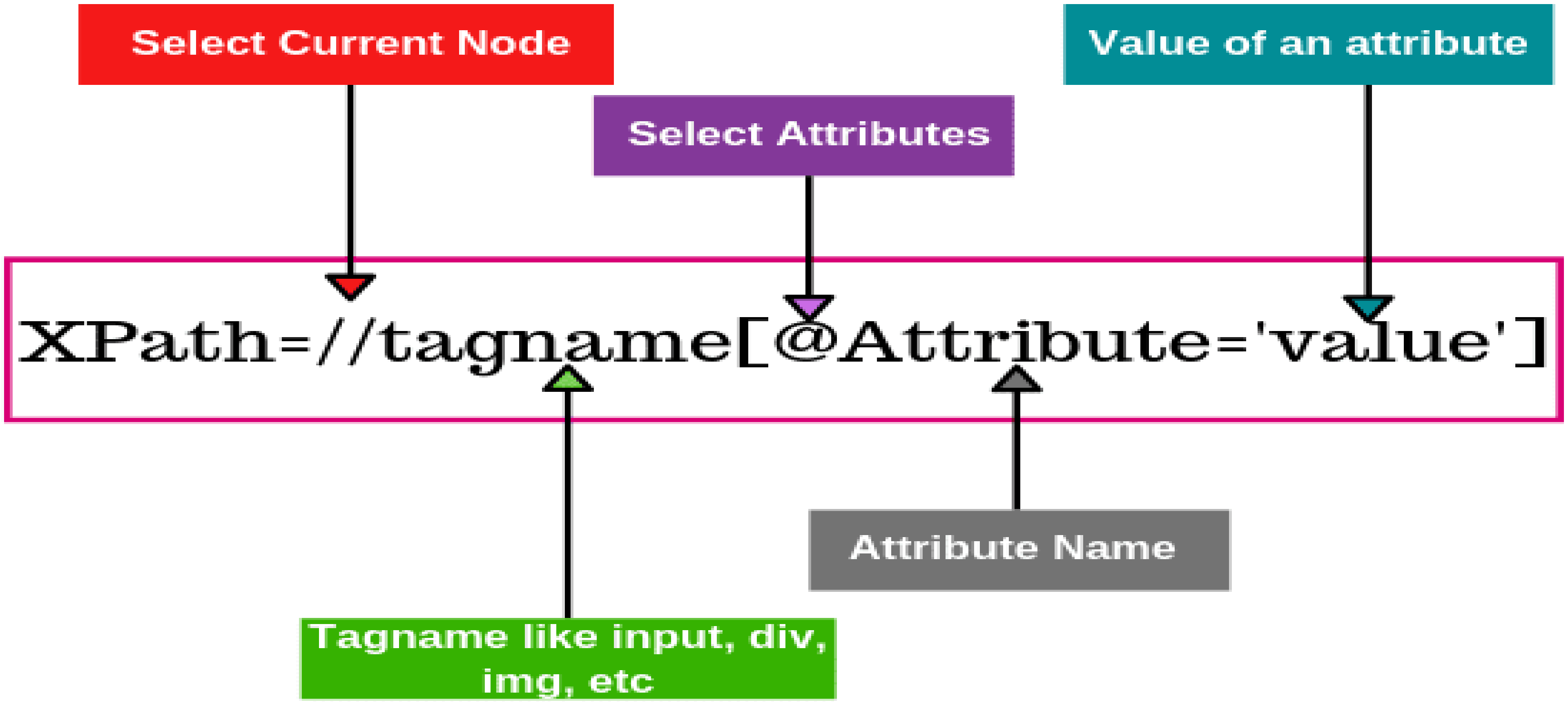


Fig: XPath Syntax and its Terminology

How to create X-Path/syntax

1. General syntax.//Tagname[@AttributeName=‘value’]
2. Finding element driver.
3. X-Path Example .//tag_name[@Attribute_name=“Value of attribute”]
4. Absolute X-Path.

/html/body/div[1]/div[3]/div/format/div[3]/div

5.Relative X-Path

//input[@id='username']

Code/Example

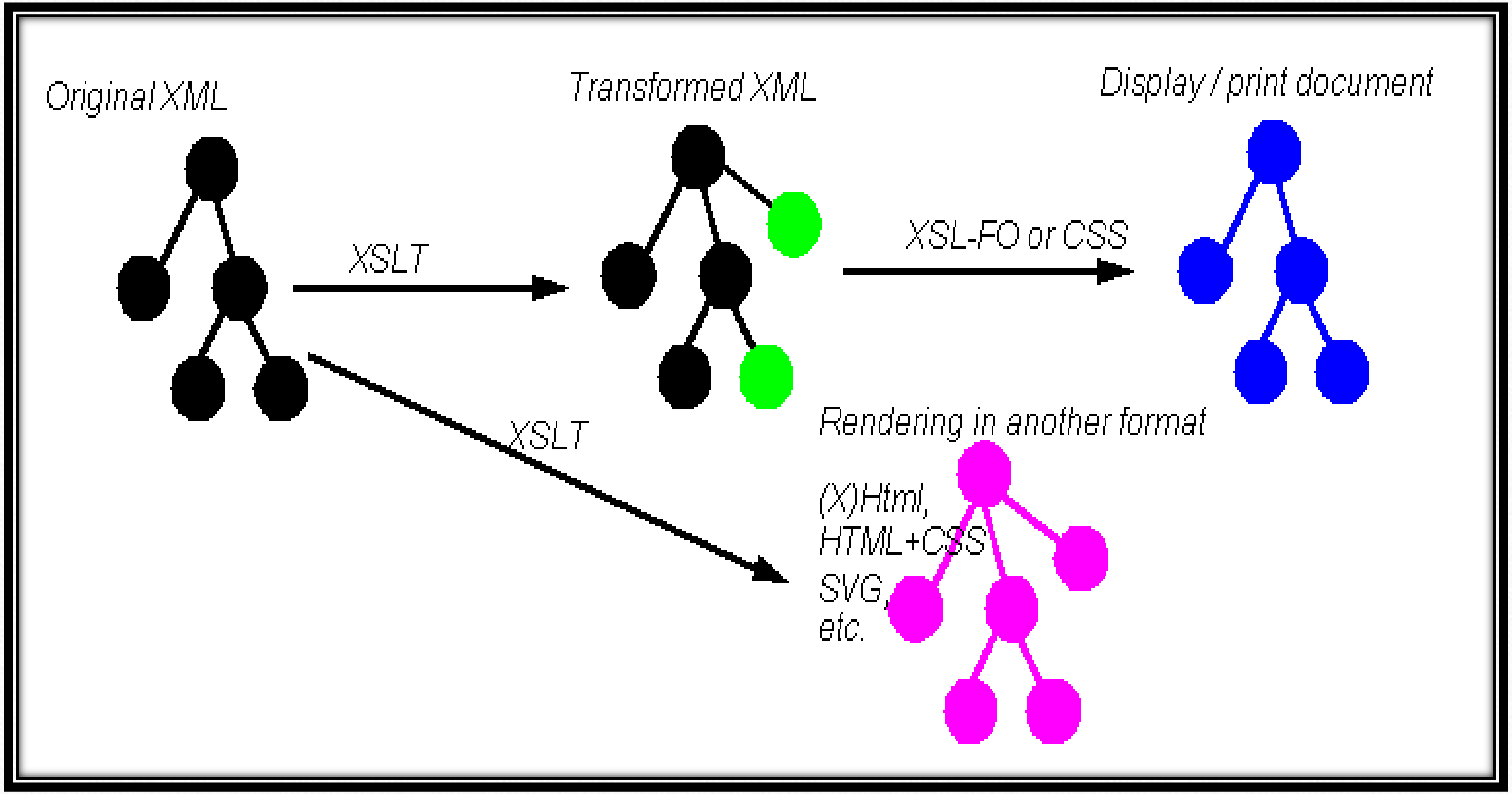
```
<bookstore>
<book category=“cooking”>
<title lang=“en”>Everyday Italian</title>
<author>Giada De Laurentils</author>
<year>2005</year>
<price>30.00</price>
</book>
<book category=“children”>
<title lang=“en”>Harry Potter</title>
<author>JK Rowling</author>
<year>2005</year>
<price>29.99</price>
</book>
</bookstore>
```

Features of X-Path

- X-Path includes over 200 built-in functions
- There are functions for strings values, numeric values, Booleans, data and time comparison, node manipulation etc
- Today X-Path can also be used in Java Script, java, XML scheme, PHP, Python and lots of other languages

Introduction to XSL-FO

- XSL-FO stands for Extensible Stylesheet Language Formatting Objects
- Styling is both about transforming and formatting information
- XSL-FO is an XML based markup language describing the formatting of XML data for output to screen, paper or other media
- A unified presentation language which stores both the content of the document and the documents formatting information


Original XML
Transformed XML
Display / print document
XSLT
XSL-FO or CSS
XSLT
Rendering in another format
(X)Html,
HTML+CSS
SVG,
etc.

How to create XSL-FO/Syntax

1. The first section contains the <fo:root> and <fo:layout-master-set>XSL FO tags that hold general page layout information for page height, page width, and page margins
2. The second section contains the XSL FO <fo:page-sequence>tag that holds the individual content elements

Code/Example

```
<?xml version="1.1" encoding="utf-8"?>
<fo:root xmlns:fo="http://www.w3.org/1999/XSL/Format">
 <fo:layout-master-set>
  <fo:simple-page-master master-name="my_page"
margin="0.5in">
   <fo:region-body/>
  </fo:simple-page-master>
 </fo:layout-master-set>
 <fo:page-sequence master-reference="my_page">
  <fo:flow flow-name="xsl-region-body">
   <fo:block>Hello world!</fo:block>
  </fo:flow>
 </fo:page-sequence>
</fo:root>
```

In this example:

fo:root is the root element of an XSL-FO document.
fo:layout-master-set defines the layout masters (templates) that can be used throughout the document.
fo:simple-page-master defines a simple page layout with regions for body content, headers, and footers.
fo:region-body, fo:region-before, and fo:region-after define the regions of the page.
fo:page-sequence defines the sequence of pages using a specific layout master.
fo:flow defines the main content area where text and other elements are placed.
fo:block is a block-level element that contains text or other block-level elements.

Features of XSL FO

- XSL-FO has properties to control hyphenation, insert leaders
- specify the number of columns on a page
- determine where page breaks occur and which paragraphs must be kept together.


Thank you